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ASSIGNMENT -1

Subject : Computer Network

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Submitted by :

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Ques 2 : Write the difference between hub, switch, and router in tabular format.
Explain why router is the most intelligent and hub is the least intelligent device.

Ans :

Difference Between Hub, Switch, and Router

A Hub, Switch, and Router are networking devices used to connect computers and other devices. However, they differ in the way they handle and forward data.

Basis	Hub	Switch	Router
OSI Layer	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Main Purpose	Connects multiple devices in a network	Connects devices and forwards data to the intended device	Connects different networks and routes data between them
Address Used	Does not use an address to make forwarding decisions	Uses MAC addresses	Uses IP addresses
Data Unit	Bits	Frames	Packets
Data Forwarding	Sends data to all connected devices	Sends data only to the destination device/port when the MAC address is known	Selects the appropriate route to reach the destination network
Collision	High possibility of collisions	Reduces collisions by providing separate collision domains	Generally separates networks and collision domains
Broadcast Domain	One broadcast domain	Normally one broadcast domain per VLAN	Separates broadcast domains by default
Traffic Management	Very limited	Better traffic management	Advanced traffic management and routing
Learning Ability	Cannot learn device addresses	Learns MAC addresses and builds a MAC address table	Maintains routing information and selects routes
Communication	Mainly supports simple network connectivity	Efficient communication within a local network	Enables communication between different networks
Security	Very low	Better than a hub	Higher, as filtering and access control

Basis	Hub	Switch	Router
Intelligence	Least intelligent	More intelligent than a hub	Most intelligent among the three
Example Use	Basic/legacy LAN connectivity	Connecting computers in a LAN	Connecting a LAN to other networks or the Internet

Why is a Router the Most Intelligent?

A router is considered the most intelligent of these three devices because it can examine IP addresses and determine the best path for forwarding packets between different networks. It maintains routing information and can make decisions based on the destination network.

Routers can also perform additional functions such as packet filtering, network address translation (NAT), traffic management, and connecting different types of networks. Therefore, they require more decision-making than hubs and switches.

Why is a Hub the Least Intelligent?

A hub is considered the least intelligent because it does not analyze the destination of the data. When a hub receives a signal, it simply broadcasts it to all the connected devices.

It does not maintain a MAC address table, does not determine the destination device, and does not make routing decisions. As a result, a hub performs only basic signal transmission and has very limited functionality.

Conclusion

In simple terms:

Hub → Broadcasts data to everyone

Switch → Sends data to the correct device within a network

Router → Determines where data should go between different networks

Therefore, the hub has the least intelligence, while the router has the highest level of decision-making capability among the three.

Ques 2 : What are random access protocols ? why do we need them ? explain each category in detail with diagram and example . Also explain diff between CSMA/CD & CSMA/CA

Ans :

Random Access Protocols are MAC (Medium Access Control) protocols used when multiple devices share the same communication channel.

In a shared network, several devices may want to transmit data at the same time. If two or more devices transmit simultaneously, their signals may interfere with each other and cause a collision.

Random access protocols provide rules that help devices decide when they can transmit data and how to handle a collision if it occurs.

Random access protocols are needed because:

- Multiple devices may use the same communication medium.
- Two devices can try to transmit at the same time.

- Simultaneous transmission can result in collisions.
- They help devices share the channel efficiently.
- They reduce unnecessary retransmissions and improve network performance.

For example, imagine several students trying to speak through a single microphone. If everyone speaks at once, nobody can understand the conversation. A random access protocol provides rules for deciding when each student should speak.

The major types of random access protocols are:

1. ALOHA
2. CSMA (Carrier Sense Multiple Access)
3. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)
4. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

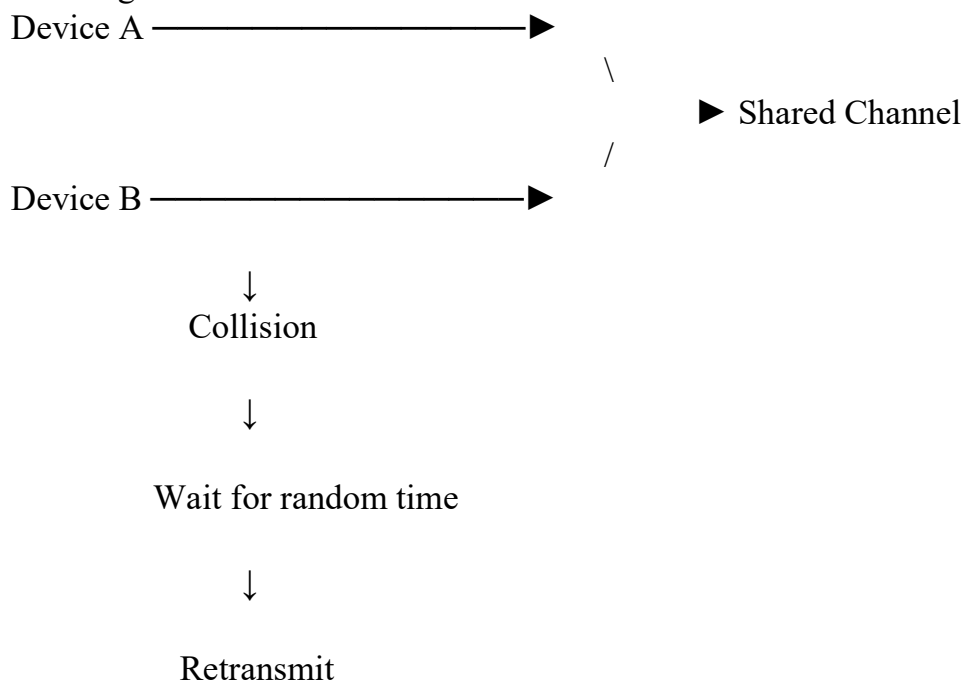
ALOHA

ALOHA is one of the simplest random access protocols.

In ALOHA, a device transmits data whenever it has data to send. It does not first check whether the communication channel is free.

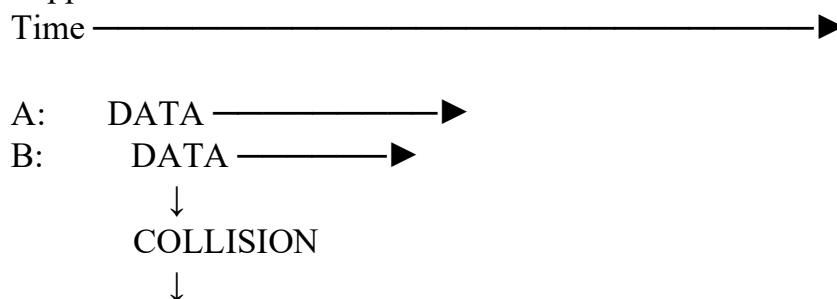
If two devices transmit at approximately the same time, a collision occurs. The devices then wait for a random amount of time and retransmit.

Working of ALOHA



Example

Suppose Device A and Device B both have data to send.



Random waiting time



A: DATA →

ALOHA is simple but can result in many collisions because devices do not check the channel before transmitting.

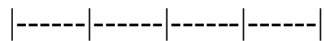
Types of ALOHA

There are two main types:

- Pure ALOHA
- Slotted ALOHA

Pure ALOHA

A device can transmit at any time when it has data.



A transmits

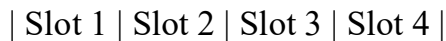
B transmits



Collision

Slotted ALOHA

Time is divided into fixed slots. A device can start transmission only at the beginning of a time slot.



Transmission

Slotted ALOHA reduces the probability of collisions compared with Pure ALOHA

CSMA

CSMA stands for Carrier Sense Multiple Access.

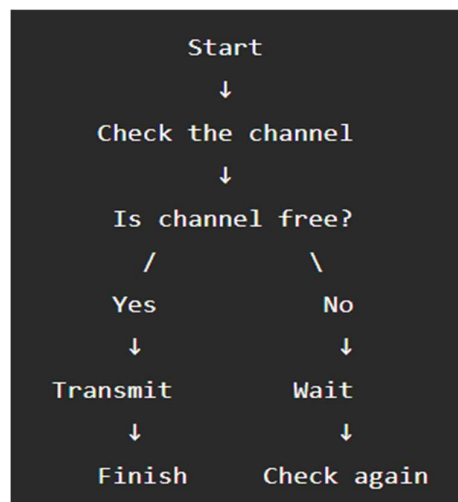
Unlike ALOHA, a device using CSMA first listens to the communication channel before transmitting.

The basic idea is:

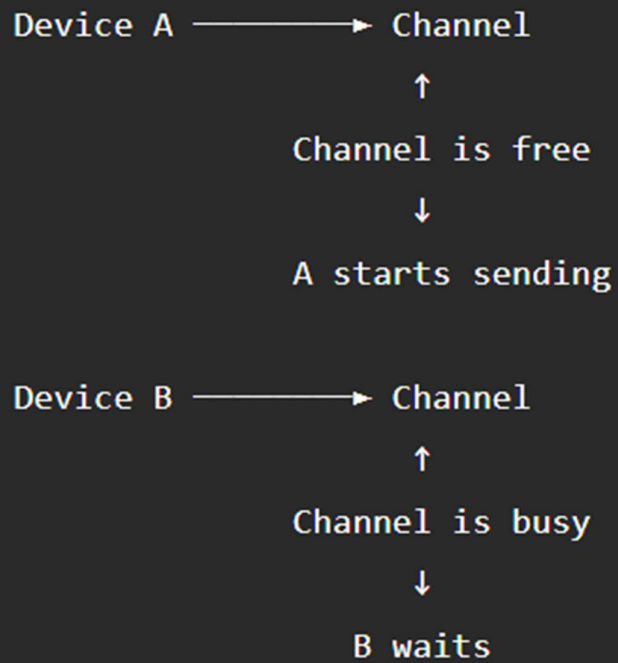
Listen before you transmit.

If the channel is free, the device can transmit. If the channel is busy, the device waits.

Working of CSMA



Example



CSMA/CD

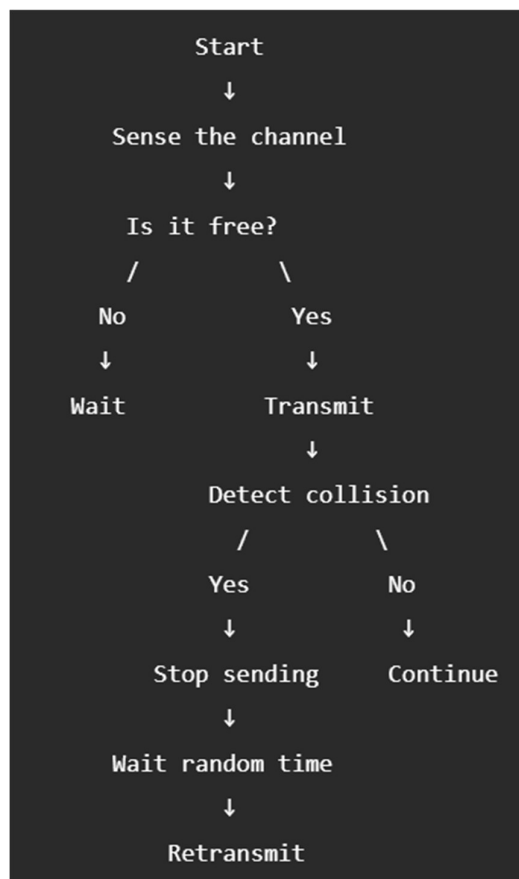
CSMA/CD stands for:

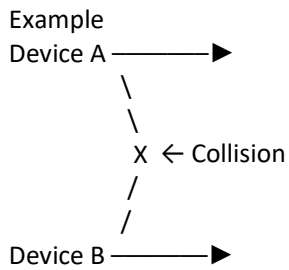
Carrier Sense Multiple Access with Collision Detection

It is an improvement over CSMA. A device checks whether the channel is free before transmission and continues monitoring the channel while transmitting.

If a collision is detected, transmission is stopped and the device waits for a random period before trying again.

Working of CSMA/CD





Both devices stop



Wait for random time



One device retransmits

CSMA/CD was commonly associated with traditional shared, half-duplex Ethernet networks. Modern switched full-duplex Ethernet normally does not experience collisions, so CSMA/CD is generally not required there.

CSMA/CA

CSMA/CA stands for:

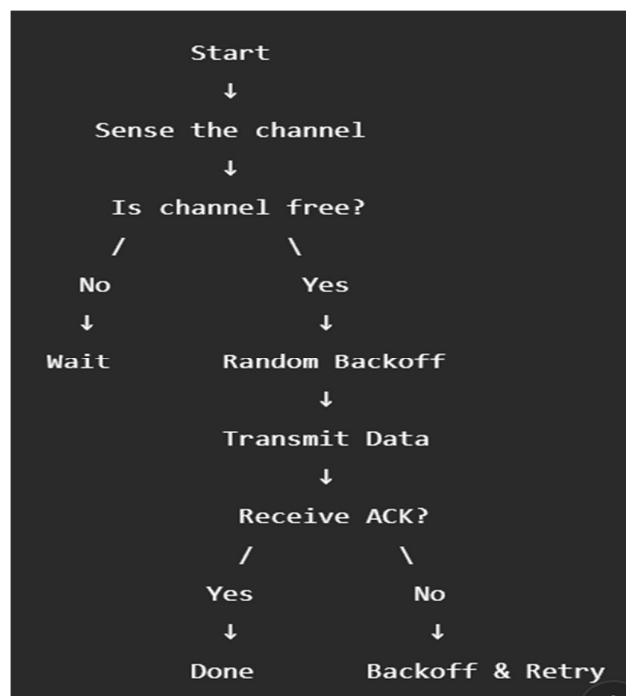
Carrier Sense Multiple Access with Collision Avoidance

It is mainly associated with wireless networks such as Wi-Fi.

In a wireless network, detecting a collision while transmitting is difficult. Therefore, CSMA/CA tries to avoid collisions before they happen.

Basic Working

1. The device checks whether the channel is free.
2. If the channel is busy, it waits.
3. If the channel is free, the device waits for a suitable period.
4. It selects a random backoff time.
5. When the backoff counter reaches zero, it transmits.
6. The receiver may send an acknowledgement (ACK).
7. If an ACK is not received, the sender assumes that the transmission may have failed and tries again.



Difference Between CSMA/CD and CSMA/CA

Feature	CSMA/CD	CSMA/CA
Full Form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Main Idea	Detect collisions after they occur	Try to avoid collisions before they occur
Commonly Used In	Traditional shared Ethernet	Wireless LAN/Wi-Fi
Collision Handling	Detects a collision during transmission	Uses waiting and backoff to reduce the chance of collision
Transmission	Device starts transmitting after sensing the channel	Device senses channel and generally uses a random backoff before transmission
ACK	Not normally required for collision detection	ACKs are commonly used in Wi-Fi
Collision Detection	Possible in traditional wired Ethernet	Difficult during wireless transmission
Main Technique	Collision detection	Collision avoidance
Example	Older shared Ethernet	IEEE 802.11 Wi-Fi

Comparison of Random Access Protocols

Protocol	Basic Principle	Collision Handling	Example
Pure ALOHA	Transmit whenever data is available	Retransmit after random waiting	Early shared networks
Slotted ALOHA	Transmit only at the beginning of a time slot	Retransmit after collision	Satellite/shared communication
CSMA	Listen before transmitting	Collision can still occur	Shared networks
CSMA/CD	Listen, transmit, and detect collisions	Stop and retransmit after random backoff	Traditional Ethernet
CSMA/CA	Listen and attempt to avoid collisions	Wait/backoff and retry	Wi-Fi

Conclusion

Random access protocols are important for controlling access to a shared communication channel. They prevent devices from transmitting without any coordination and provide methods for dealing with collisions.

ALOHA allows devices to transmit with very little checking, while CSMA first senses the channel. CSMA/CD detects collisions during transmission and is associated with traditional shared Ethernet, whereas CSMA/CA attempts to avoid collisions and is mainly used in wireless networks.

In short:

ALOHA → Transmit first

CSMA → Listen before transmitting

CSMA/CD → Listen + Detect collision

CSMA/CA → Listen + Avoid collision